Model Accuracy Report

HR Analytics — Employee Attrition Prediction

1. Overview

Three classification models were trained to predict employee attrition using the IBM HR Analytics dataset (1,470 records, stratified 80/20 train-test split). Because the dataset is imbalanced (~16% attrition rate), accuracy alone is not a reliable measure of model quality — precision, recall, and F1-score on the attrition (minority) class were used as the primary evaluation metrics.

2. Model Comparison

Model	Accuracy	Precision (Attrition)	Recall (Attrition)	F1-Score
Logistic Regression	86.1%	0.62	0.34	0.44
Decision Tree	83.3%	0.44	0.17	0.25
Random Forest	83.3%	0.42	0.11	0.17

Logistic Regression was selected as the final model because it achieved the highest accuracy (86.1%) and the highest recall (0.34) and F1-score (0.44) for the attrition class among the three evaluated models. While Decision Tree and Random Forest show similar overall accuracy, both perform substantially worse at identifying employees who actually leave (lower recall and F1-score on the attrition class), making them less useful for a proactive retention strategy.

3. Confusion Matrix — Logistic Regression (Final Model)

Logistic Regression	Predicted: No	Predicted: Yes
Actual: No	TN=237	FP=10
Actual: Yes	FN=31	TP=16

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4. Confusion Matrix — Decision Tree

Decision Tree	Predicted: No	Predicted: Yes
Actual: No	TN=237	FP=10
Actual: Yes	FN=39	TP=8

5. Confusion Matrix — Random Forest

Random Forest	Predicted: No	Predicted: Yes
Actual: No	TN=240	FP=7
Actual: Yes	FN=42	TP=5

6. Interpretation

- Recall on the attrition class matters most: it measures how many actual leavers the model correctly catches, which is the core business goal.
- Logistic Regression's higher recall (0.34) means it identifies roughly 1 in 3 employees who will leave — modest, but meaningfully better than the tree-based models.
- False negatives represent employees who actually leave but are predicted to stay, which can result in missed opportunities for proactive retention efforts.
- Future improvement: applying class-balancing techniques such as class_weight='balanced' or SMOTE oversampling could improve recall further.